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$$x_n = 1 \quad s_n = 31 \quad q = 0,5$$

$$x_1 = \left(\frac{1}{2}\right)^{1-n}$$

$$s_n = x_1 \frac{1 - q^n}{1 - q}$$

$$\Leftrightarrow 31 = \left(\frac{1}{2}\right)^{1-n} \cdot \frac{\left(1 - \left(\frac{1}{2}\right)^n\right)}{\frac{1}{2}}$$

$$\Leftrightarrow 16 = \left(\frac{1}{2}\right)^{1-n}$$

$$\Leftrightarrow 2^4 = 2^{n-1}$$

$$\Leftrightarrow n = 5$$

References